

MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 4

1. *Find the inverses of the following matrices or show that the matrix is not invertible:*

(1) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

(3) $\begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

(1) Here we use the Gauss-Jordan algorithm:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

We conclude that the inverse is $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

(2) Once again, we use the Gauss-Jordan algorithm to calculate the inverse of the matrix:

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/2 & -1 & 1/2 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to \frac{2}{3}R_2} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & -2/3 & 1/3 & 2/3 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + R_2} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & -2/3 & 1/3 & 2/3 & 0 \\ 0 & 0 & 4/3 & 1/3 & 2/3 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to \frac{3}{4}R_3} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & -2/3 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & 1/2 & 3/4 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + \frac{2}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 1 & 1/2 \\ 0 & 0 & 1 & 1/4 & 1/2 & 3/4 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 + \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & 1/2 & 1/4 \\ 0 & 1 & 0 & 1/2 & 1 & 1/2 \\ 0 & 0 & 1 & 1/4 & 1/2 & 3/4 \end{array}\right]$$

We conclude that the inverse is $A^{-1} = \begin{bmatrix} 3/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \\ 1/4 & 1/2 & 3/4 \end{bmatrix}$.

(3)

$$\left[\begin{array}{cccc|cccc} 2 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R3 \leftrightarrow R4} \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R3 \to R3 - R4, R1 \to R1 + R4} \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R2 \leftrightarrow R1} \left[\begin{array}{cccc|cccc} 1 & -1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R2 \to R2 - 2R1} \left[\begin{array}{cccc|cccc} 1 & -1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 3 & -2 & 0 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{R3\to R3-R1}\left[\begin{array}{cccc|cccc}1&-1&1&0&0&1&0&0\\0&3&-2&0&1&-2&1&0\\0&2&0&0&0&-1&-1&1\\0&0&0&1&0&0&1&0\end{array}\right]\xrightarrow{R2\to R2/3}\left[\begin{array}{cccc|cccc}1&-1&1&0&0&1&0&0\\0&1&-2/3&0&1/3&-2/3&1/3&0\\0&2&0&0&0&-1&-1&1\\0&0&0&1&0&0&1&0\end{array}\right]$$

$$\xrightarrow{R1\to R1+R2}\left[\begin{array}{cccc|cccc}1&0&1/3&0&1/3&1/3&1/3&0\\0&1&-2/3&0&1/3&-2/3&1/3&0\\0&2&0&0&0&-1&-1&1\\0&0&0&1&0&0&1&0\end{array}\right]$$

$$\xrightarrow{R3\to R3-2R2}\left[\begin{array}{cccc|cccc}1&0&1/3&0&1/3&1/3&1/3&0\\0&1&-2/3&0&1/3&-2/3&1/3&0\\0&0&4/3&0&-2/3&1/3&-5/3&1\\0&0&0&1&0&0&1&0\end{array}\right]$$

$$\xrightarrow{R3\to 3R3/4}\left[\begin{array}{cccc|cccc}1&0&1/3&0&1/3&1/3&1/3&0\\0&1&-2/3&0&1/3&-2/3&1/3&0\\0&0&1&0&-2/4&1/4&-5/4&3/4\\0&0&0&1&0&0&1&0\end{array}\right]$$

$$\xrightarrow{R2\to R2+2/3R3}\left[\begin{array}{cccc|cccc}1&0&1/3&0&1/3&1/3&1/3&0\\0&1&0&0&0&-1/2&-1/2&1/2\\0&0&1&0&-2/4&1/4&-5/4&3/4\\0&0&0&1&0&0&1&0\end{array}\right]$$

$$\xrightarrow{R1\to R1-1/3R3}\left[\begin{array}{cccc|cccc}1&0&0&0&1/2&1/4&3/4&-1/4\\0&1&0&0&0&-1/2&-1/2&1/2\\0&0&1&0&-2/4&1/4&-5/4&3/4\\0&0&0&1&0&0&1&0\end{array}\right].$$

So the inverse matrix is:

$$\left[\begin{array}{cccc}1/2&1/4&3/4&-1/4\\0&-1/2&-1/2&1/2\\-2/4&1/4&-5/4&3/4\\0&0&1&0\end{array}\right]$$

(4)

$$\left[\begin{array}{ccc|ccc}2&1&2&1&0&0\\4&2&1&0&1&0\\2&1&1&0&0&1\end{array}\right]\xrightarrow{R2\to R2-2R1}\left[\begin{array}{ccc|ccc}2&1&2&1&0&0\\0&0&-3&-2&1&0\\2&1&1&0&0&1\end{array}\right]$$

$$\xrightarrow{R3\to R3-R1}\left[\begin{array}{ccc|ccc}2&1&2&1&0&0\\0&0&-3&-2&1&0\\0&0&-1&-1&0&1\end{array}\right]\xrightarrow{R3\to R3-R2/3}\left[\begin{array}{ccc|ccc}2&1&2&1&0&0\\0&0&-3&-2&1&0\\0&0&0&-1/3&-1/3&1\end{array}\right]$$

We get a row of zeros in the left hand side, so it is not possible to transform left hand side to the identity matrix. Thus, A is not invertible.

2. *Determine whether each of the following matrices are invertible. Justify your answer. It is not necessary to find the inverse if it exists (but you can if it helps you determine if it is invertible).*

(1) $\begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$

(2) $\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix}$

(3) $\begin{bmatrix} 0 & 3 & -5 \\ 1 & 0 & 2 \\ -4 & -9 & 7 \end{bmatrix}$

(4) $\begin{bmatrix} -1 & 3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix}$

In each case, it is sufficient to check if an REF of the matrix has the full number of pivots. This will imply that if we were to do the Gauss Jordan algorithm, we would succeed in transforming the right-hand side into the identity matrix, thus producing an inverse.

(1) Taking the matrix to a REF:

$$\begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix} \xrightarrow{R_2 \to R_2 + \frac{3}{5}R_1} \begin{bmatrix} 5 & 7 \\ 0 & -9/5 \end{bmatrix}$$

There are two pivots in the above REF, thus the matrix is invertible.

(2) Again, taking the matrix to a REF:

$$\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix} \xrightarrow{R_2 \to R_2 + \frac{3}{5}R_1} \begin{bmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - \frac{8}{5}R_1} \begin{bmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 5 & -1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + \frac{5}{7}R_2} \begin{bmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

There are three pivots in the above REF, thus the matrix is invertible.

(3) Again, taking the matrix to a REF:

$$\begin{bmatrix} 0 & 3 & -5 \\ 1 & 0 & 2 \\ -4 & -9 & 7 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ -4 & -9 & 7 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 4R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ 0 & -9 & 15 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 3R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

We see that there are only two pivots in the above REF, thus the matrix is not invertible.

(4) Again, taking the matrix to a REF:

$$\begin{bmatrix} -1 & 3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 + 3R_1} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 14 & 8 & 0 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_1} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 14 & 8 & 0 \\ 0 & -12 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2 \to \frac{1}{14}R_2} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 1 & 4/7 & 0 \\ 0 & -12 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 12R_2} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 1 & 4/7 & 0 \\ 0 & 0 & 68/7 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{R_4 \to R_4 + R_2} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 1 & 4/7 & 0 \\ 0 & 0 & 68/7 & 0 \\ 0 & 0 & 18/7 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to \frac{7}{68} R_3} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 1 & 4/7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 18/7 & 1 \end{bmatrix} \xrightarrow{R_4 \to R_4 - \frac{18}{7} R_3} \begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & 1 & 4/7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

There are four pivots and thus this matrix is invertible.

3. *Answer the following true-false questions. Justify your answers! A and B are arbitrary $n \times n$ square matrices.*

(1) *If A is invertible then $A\mathbf{x} = \mathbf{0}$ has exactly one solution, $\mathbf{x} = \mathbf{0}$.*
(2) *If A is invertible, then AB is also invertible.*
(3) *If A and B are invertible, then $A + B$ is also invertible.*
(4) *If A is invertible, then the reduced echelon form of A is equal to I.*
(5) *If A is invertible and has an LU decomposition $A = LU$, then both L and U are invertible.*
(6) *If A is not invertible and has an LU decomposition $A = LU$, then neither L nor U are invertible.*
(7) *If A has an LU factorization, then this factorization is unique.*

(1) True, $A\mathbf{x} = \mathbf{b}$ has *at least* one solution, $\mathbf{x} = A^{-1}\mathbf{b}$. To see that $A\mathbf{x} = \mathbf{x}$ has *at most* one solution, suppose that $\mathbf{x}_1, \mathbf{x}_2$ are solutions, i.e., $A\mathbf{x}_1 = A\mathbf{x}_2 = \mathbf{b}$. Multiplying through by A^{-1} we see that $\mathbf{x}_1 = \mathbf{x}_2 = A^{-1}\mathbf{b}$, i.e., all solutions *must* be equal to $A^{-1}\mathbf{b}$ and to each other. Hence there is also *at most* one solution.
(2) False, let $A = I$ and $B = 0$.
(3) False, let $A = I$ and $B = -I$.
(4) True, the reduced echelon form of an $n \times n$ invertible matrix A will have exactly n pivots, for which the entries above and below are zero, i.e. the reduced echelon form is the identity I.
(5) True. If the product of two matrices is invertible, then each one of the factors is invertible.
(6) False. Let $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Since $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ are both LU decompositions, in the first case U is invertible and in the second case L is invertible.
(7) False, if $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then $L = U = A$ is an LU decomposition, as well as $L = A, U = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$.

4. *Let*

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}.$$

*Such a matrix called **tridiagonal matrix**. Band matrices often appear in applications (see the next exercise) and they are one reason LU-decomposition is so important.*

(1) *Find the LU-decomposition of A.*
(2) *Determine A^{-1}.*

Given a vector $\boldsymbol{b}$ in $\mathbb{R}^5$, suppose we want to solve $A\boldsymbol{x} = \boldsymbol{b}$. In many applications, we probably don't want to do this for a single $\boldsymbol{b}$, but for many (maybe a million) different vectors $\boldsymbol{b}$. One way for finding $\boldsymbol{x}$ is to solve the following two linear systems which arise from the LU-decomposition of A:

$$L\boldsymbol{c} = \boldsymbol{b}, \text{ and } U\boldsymbol{x} = \boldsymbol{c}.$$

Another way is to calculate A^{-1}, and then we would get $\boldsymbol{u}$ by simply multiplying $\boldsymbol{b}$ by A^{-1}. Which way is more efficient?

(3) *Given the LU-decomposition of A, count the numbers of operations (addition and multiplication of real numbers) needed to find $\boldsymbol{x}$ using first method?*

(4) *Given the inverse of A, count the number of operations needed to calculate $A^{-1}\boldsymbol{b}$?*

(5) *Suppose A is not a 5×5 band matrix, but a 1000×1000 band matrix? Which of the two methods would you use?*

(1) Doing the row operations $R_2 \to R_2 + \frac{1}{2}R_1$, $R_3 \to R_3 + \frac{2}{3}R_2$,$R_4 \to R_4 + \frac{3}{4}R_3$, $R_5 \to R_5 + \frac{4}{5}R_4$, we get to the echelon form.

$$U = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -1 & 0 \\ 0 & 0 & 0 & \frac{5}{4} & -1 \\ 0 & 0 & 0 & 0 & \frac{6}{5} \end{bmatrix}.$$

We get L by applying the inverses of these row operations, in reverse operation to the identity matrix. This then gives us that

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 & 0 \\ 0 & -\frac{2}{3} & 1 & 0 & 0 \\ 0 & 0 & -\frac{3}{4} & 1 & 0 \\ 0 & 0 & 0 & -\frac{4}{5} & 1 \end{bmatrix}$$

(2) A comptuer calculation (or using Gauss Jordan) gives that the inverse of this matrix is

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 5 & 4 & 3 & 2 & 1 \\ 4 & 8 & 6 & 4 & 2 \\ 3 & 6 & 9 & 6 & 3 \\ 2 & 4 & 6 & 8 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

(3) To understand the complexity of solving $Lc = b$, we need to know how many row operations to get L to the identity matrix. To get the second row right, we need to add $\frac{1}{2}$ of the first row to the second row, which requires four operations (two each for the 1 in the first column, and two for the augmented portion). Similarly for the third row, we need to add $\frac{2}{3}$ of the second row to the third row. Continuing this pattern, we need 16 operations to solve $Lc = b$.

We now consider how many operations to calculate $Ux = c$. We first multiply the fifth row by $\frac{5}{6}$, which is two operations. Then we add the fifth row to the fourth,

which is two operations. Continuing like this, we get that this takes 18 operations. Thus all together this solution takes 34 operations.

It is worth noting that the number of operations here scale linearly as the size of the matrix grows.

(4) We consider $A^{-1}b$. To calculate the each entry of this, we need to do five multiplications and five additions. This then takes 50 operations. Note that this scales with the square of the size of A.

(5) Looking at how these scale (linearly versus quadratically), the LU decomposition is the superior solution method.